

Fig. 13: Marx bank power supply arrangement

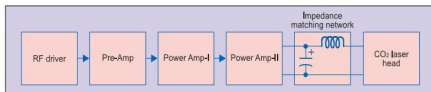


Fig. 14: Power supply for RF-excited carbon dioxide laser



Fig. 15: Carbon dioxide laser power supplies

tion. This means, if each capacitor in the bank were charged to a voltage V , then the total voltage available across the tube for discharge operation would be nV , n being the number of capacitors.

Fig. 13 shows the Marx bank power supply arrangement where the switches are open during the charging. They are triggered simultaneously to the closed position to initiate discharge. Though the Marx bank shown here uses four stages, Marx bank systems with higher number of stages are also used. Marx banks are particularly suitable for producing fast-rising pulses.

Power supply of an RF-excited CO_2 laser comprises an RF source and an impedance-matching network (Fig. 14). The RF source may further be split up into an RF oscillator and a cascade arrangement of RF amplifiers, depending upon the output power

delivering capability. Frequency of operation is in the range of 50MHz to 150MHz. RF power is fed to the laser cavity through impedance-matching network. The laser cavity's mechanical structure combined with RF discharge can be considered as an electrical load comprising complex impedance. Power transfer efficiency from the RF source to the discharge is a key operational feature of the laser that is made more complicated by the fact that the discharge impedance is power dependent.

CO_2 laser power supplies too are commercially available both as bench-top laboratory systems as well as compact versions for OEM market (Fig. 15).

Frequency stabilisation of lasers

Carbon dioxide and helium-neon lasers have Doppler broadened gain

versus frequency curves, which is 60MHz in the case of carbon dioxide laser and 1400MHz for helium-neon laser emitting at 632.8nm. With active stabilisation techniques, such as dither stabilisation, optical galvanic stabilisation and stark cell stabilisation, these lasers can be frequency stabilised up to an accuracy of $\pm 1\text{MHz}$.

Fig. 16 shows block diagram of dither stabilisation scheme. The grating allows selection of various lines within the laser tuning curve. The output of the low-frequency oscillator, when applied to the PZT, moves the grating along the axis and hence varies the output. The other input to the frequency synchronous detector is the error signal, which corresponds to the deviation in the laser output frequency from the line centre.

In fact, the magnitude and phase of the error signal determine the location of operational point with respect to the centre of the Doppler broadened gain curve. While the amplitude decides how far or close to the line centre the operational point is, the phase decides which side of the line centre it is located. Though Dither stabilisation is simple to implement, the disadvantage of this technique is the presence of frequency modulation in the output as a result of laser dithering.

Optogalvanic stabilisation technique is particularly suitable for carbon dioxide lasers. It makes use of variation in discharge impedance due to alteration in internal radiation field intensity. Fig. 17 shows a typical optogalvanic frequency stabilisation set up.

An AC signal is applied to the cavity length transducer resulting in frequency modulation of the laser and therefore modulation of the plasma-tube impedance, which is proportional to the slope of the output power versus frequency curve of the laser, is measured by exciting the plasma-tube by a high-speed current-regulated power supply and measuring the resulting variation in voltage-drop across the plasma-tube. This voltage is applied